Test Document

Author

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TABLE I
Comparative Summary of Prior Studies on Machine Learning for CKD Prediction vs. This Work

Study	Cohort & Source	Target Task	Validation	Leakage Status	Reported Findings / Critique
Almansour et al. (2019)	UCI CKD (400 records)	Binary (CKD vs. Not)	70/30 Holdout Split	Critical Leakage (S_{cr} kept)	97.5% Acc. via ANN; cannot stage severity; overoptimistic holdout.
Senan et al. (2021)	UCI CKD (400 records)	Binary	10-fold CV	Global Preprocessing	98.2% Acc. via SVM; imputation executed prior to fold splitting.
Debal & Sitote (2022)	UCI CKD (400 records)	Binary	5-fold CV	Severe Leakage	99.1% Acc. using Random Forest; exploited serum creatinine thresholding.
Chen et al. (2020)	EHR Cohort ($N = 1,200$)	5-Stage KDIGO	10-fold CV	Partial (S_{cr} included)	82.4% Acc.; predicted eGFR categories with creatinine in inputs.
Ravizza et al. (2019)	Longitudinal EHR ($N = 10,450$)	eGFR Decline (ΔeGFR)	Patient-level Split	Leakage-Free	C-index 0.79 for rapid progression; required longitudinal lab history.
Schuld et al. (2019)	Theoretical Benchmark	Synthetic Binary	Holdout	Synthetic Data	Formalized quantum kernel Hilbert space embeddings; no clinical validation.
Havlicek et al. (2019)	Superconducting QPU	2-Qubit Toy Task	Holdout	Synthetic Data	Demonstrated ZZ-feature map kernel; demonstrated vulnerability to noise.
This Work	UCI CKD ($N = 374$ eligible)	5-Class & 3-Class eGFR	Nested 5×5 CV (25 Folds)	Strictly Leakage-Free (S_{cr}, Age Excluded)	0.8204 / 0.8563 QWK; proven negative